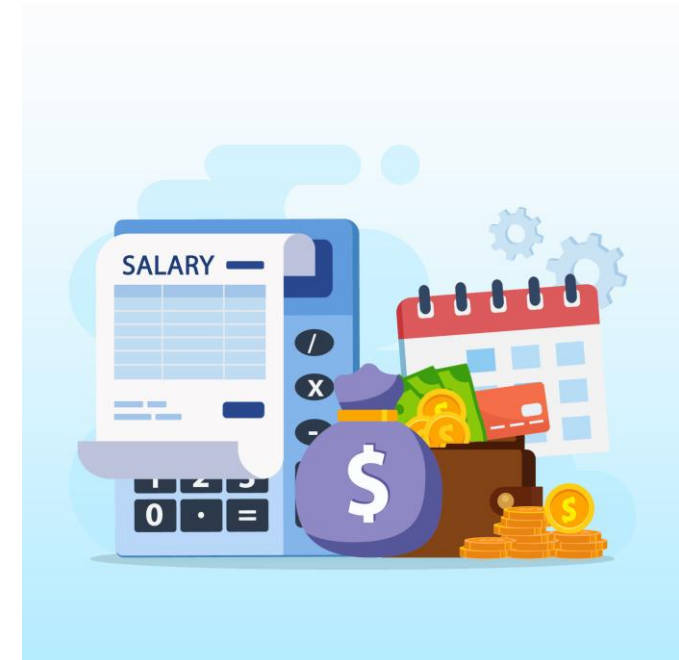
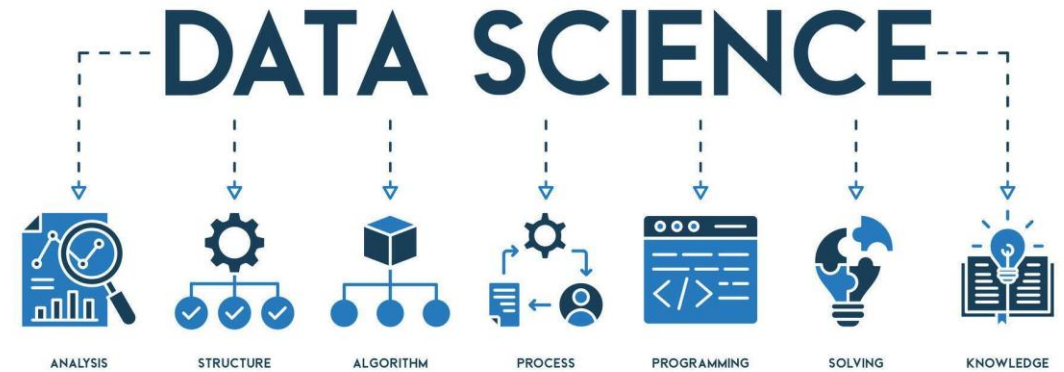


Salary Median Classification for Data Science Roles Predicting Above-Median vs. Below-Median Salaries Using Machine Learning

By **Aaron Niecestro**



Project Overview

Goal: Classify whether a data-science professional earns above or below the median salary using machine-learning models.

Why This Matters: Organizations need clear, data-driven insights to support compensation planning, pay equity, and competitive hiring.

What This Project Covers:

- Understanding the dataset and key salary-related features
- Building and comparing multiple ML classification models
- Identifying the strongest predictors of higher salary outcomes
- Selecting the best model for accuracy and interpretability
- Providing actionable insights for HR, recruiting, and workforce strategy

Outcome: A reliable, interpretable salary-classification model that highlights the factors most strongly associated with higher compensation.

Introduction

Understanding Data Science Salaries

What This Dataset Represents

- Real salary records for U.S. data-science and analytics professionals
- Includes job titles, experience levels, company size, remote work, and employment type
- Median salary category used as a stable classification target

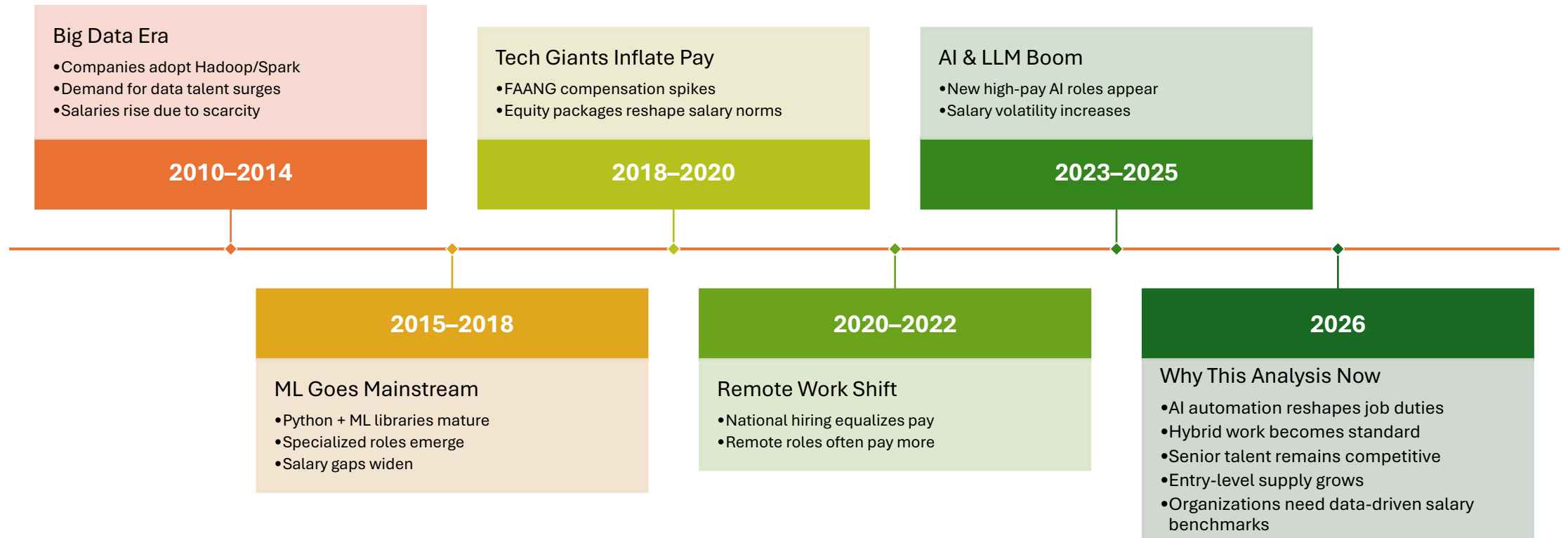
Why Salary Analysis Matters

- Salary variation is large across roles, industries, and experience levels
- Companies need data-driven compensation benchmarks
- Job seekers want clarity on what drives higher pay
- Supports pay-equity, hiring strategy, and workforce planning

Purpose of This Analysis

- Build machine-learning models to classify salaries as above or below median
- Identify the strongest predictors of higher compensation
- Provide interpretable insights for HR, recruiting, and leadership

History of Data Science Salaries — Why This Analysis Matters



Preview of Data & Analysis Approach

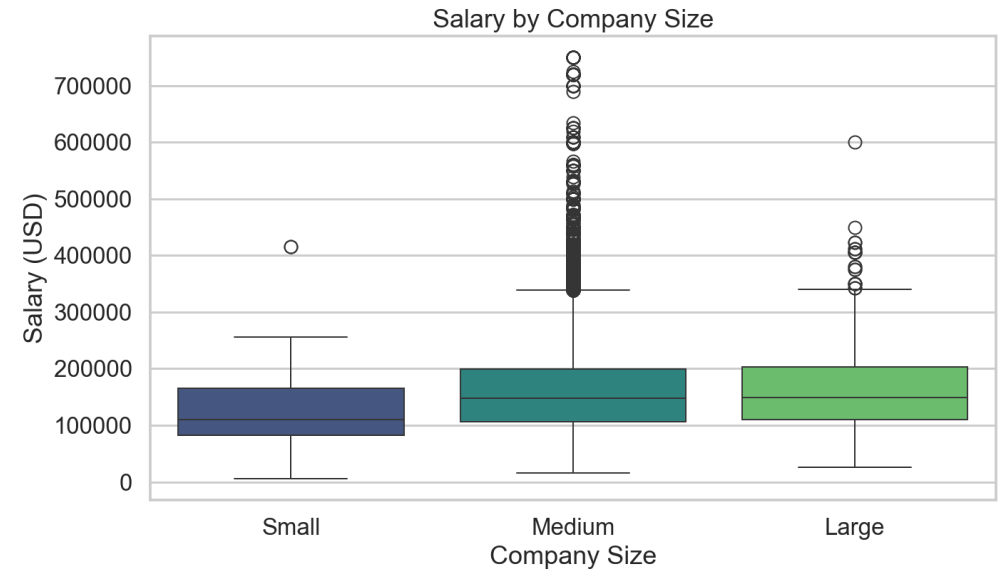
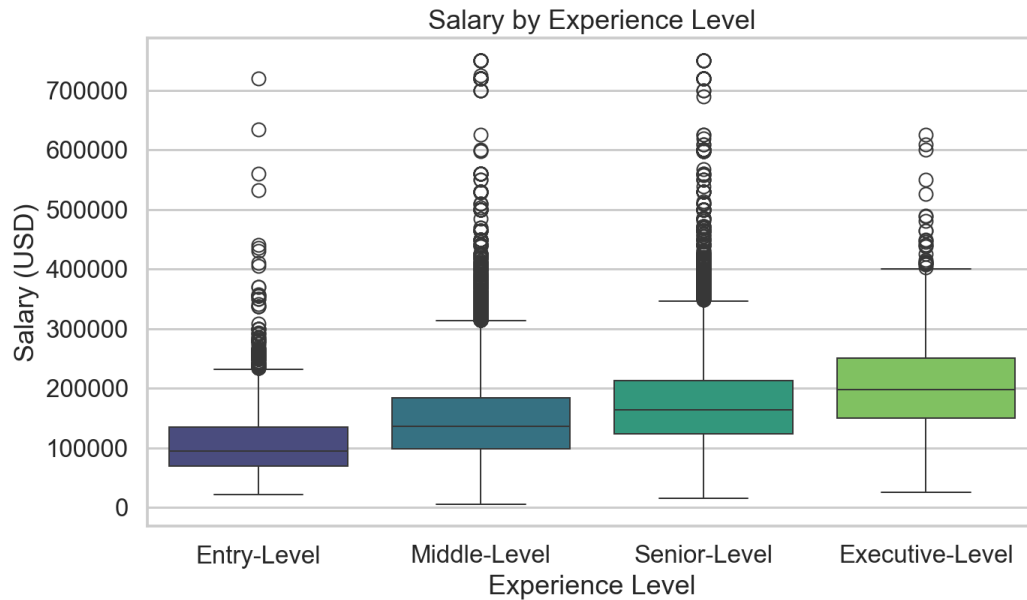
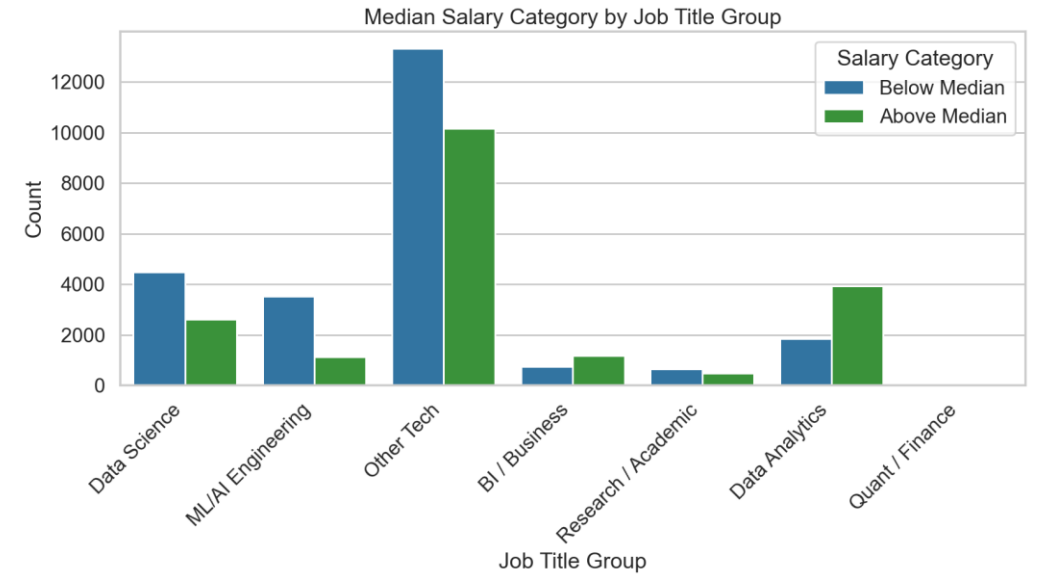
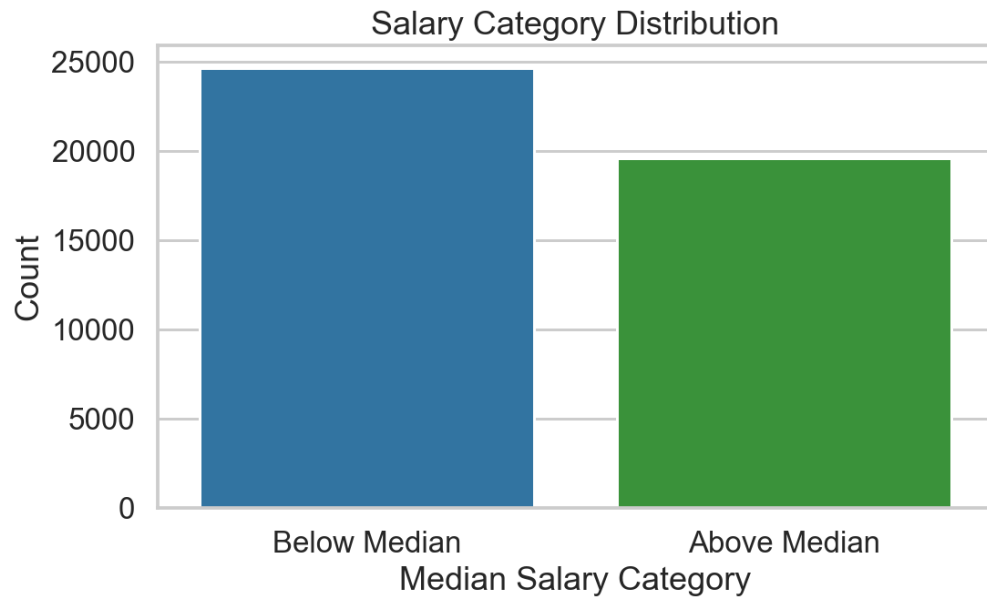
Data Description

U.S.-only subset used for consistent compensation comparison

- **Data has 44,225 observations**
- **Median Salary Category (Target)**
 - Below (Median < \$147,700)
 - Above (Median ≥ 147,700)
- **Key Feature Groups**
 - **Job & Experience Attributes**
 - Experience level
 - Job title group
 - Work year (2020–2025)
 - Data age (recency of salary record)
 - **Employment & Work Arrangement**
 - Employment type (full-time, contract, etc.)
 - Employee residence
 - Employee remote ratio (on-site, hybrid, remote)
 - Employer remote ratio (company's remote policy)
 - **Company Attributes**
 - Company size (small, medium, large)
- **Why These Features Matter**
 - These attributes capture the most important drivers of salary variation:
 - Seniority
 - Job specialization
 - Company scale
 - Remote work flexibility
 - Geographic factors

Snapshot of the Data Table

Job Title	Experience Level	Company Size	Remote Ratio	Employee Residence	Salary Median Category
Data Analytics	Entry-Level	Medium	0.0	United States	Below-Median
Other Tech	Senior-Level	Medium	0.0	United States	Above-Median
Other Tech	Middle-Level	Medium	50.0	United States	Above-Median
Other Tech	Senior-Level	Medium	0.0	United States	Above-Median
ML/AI Engineering	Senior-Level	Medium	0.0	United States	Above-Median



Insights from Data and Graphs

- This dataset contains structured salary records for U.S. data-science professionals. Each row represents one job record with attributes describing the role, company, and work arrangement.”
- Senior and specialized roles consistently fall above the median salary, while entry-level and generalist roles cluster below it.
- Job title group emerges as the #1 driver of salary in both feature importance and SHAP analysis.

Key Insights from the Data

- **1. Job Role & Seniority Drive Salary**
 - Senior and specialized roles consistently fall above the median
 - Entry-level and generalist roles cluster below the median
- **2. Experience Strongly Predicts Compensation**
 - Clear upward trend: more experience → higher salary
- **3. Company Size Matters**
 - Large companies pay more on average
 - Small companies show lower salary distributions
- **4. Remote Work Influences Pay**
 - Fully remote roles trend higher
 - On-site roles cluster below median
- **5. Median Salary Is a Stable Target**
 - Less sensitive to outliers than mean salary
 - Provides a reliable classification threshold

Modeling Approach

Goal:	Models Tested	How We Evaluated Performance	Final Model Variables Used:	Why This Matters
• To identify the most reliable salary-classification model, we evaluated a diverse set of machine-learning algorithms — from simple interpretable models to advanced nonlinear ensembles.	• Interpretable Models • Logistic Regression • Naive Bayes • Distance & Margin-Based Models • K-Nearest Neighbors • Support Vector Machines (Linear & RBF) • Ensemble Models • Decision Tree • Random Forest • Gradient Boosting • AdaBoost • Extra Trees	• Each model was trained on the same data split (70/30) and assessed using: • Accuracy — overall correctness • Precision — correctness of above-median predictions • Recall — ability to find true above-median earners • F1 Score — balance of precision and recall • Area under the Curve (AUC) — overall ranking ability (most reliable metric)	• Experience Level • Employment Type • Job Title by Group • Employee residence • Company Size • Work Calendar Year • Employer Remote Type	• Testing multiple model families ensures we select the most accurate, stable, and interpretable approach for salary classification — not just the first model that works.

How We Evaluate Model Performance

Accuracy — Overall Correctness

- What it means: How often the model is right.

Precision — Correct High-Salary Predictions

- What it means: When the model predicts “above median,” how often it’s correct.

Recall — Ability to Find High Earners

- What it means: How many real above-median earners the model successfully identifies.

F1 Score — Balanced Quality

- What it means: A balance of precision and recall — useful when classes differ.

ROC-AUC — Separation Strength

- What it means: How well the model distinguishes between high and low earners.
- Initial most important metric for this project because it measures how well the model separates above-median vs. below-median earners.

Interpretable Model

“Baseline models that show linear and probabilistic salary patterns.”

Logistic Regression

What is it?

A simple, interpretable model that predicts whether salary is above or below the median

Why It's Useful

- Provides a clear baseline for comparison
- Shows how well linear patterns explain salary outcomes
- Produces interpretable coefficients (direction + strength of influence)

Key Takeaway

- Logistic Regression gives us a clean, interpretable view of salary drivers — essential for compensation strategy and pay-equity analysis.

Why This Matters

- Helps identify which features push salary up or down
- Supports transparent, policy-driven decisions (HR, compensation, pay equity)
- Serves as the foundation for the reduced GLM model used later

Metric	Score
Accuracy	65.27%
Precision	67.39%
Recall	73.01%
F1 Score	70.09%
ROC-AUC	69.08%

Naive Bayes

What is it?

A fast, simple model that makes predictions based on probability patterns in the data.

Why It's Useful

- Extremely quick to train
- Provides a lightweight benchmark
- Helps confirm whether simple statistical patterns can separate above- vs. below-median salaries
- Useful for comparison against more complex models

Key Takeaway

- Naive Bayes provides a fast, simple benchmark but struggles with correlated features — making it less accurate than Logistic Regression and tree-based models.

Why This Matters

- Confirms that salary patterns are not purely independent
- Reinforces the need for more flexible models
- Helps validate that improvements in later models are meaningful

Metric	Score
Accuracy	56.16%
Precision	56.01%
Recall	99.45%
F1 Score	71.66%
ROC-AUC	65.67%

Distance & Margin-Based Models

“Models that capture nonlinear boundaries and local structure.”

K-Nearest Neighbors (KNN)

What is it?

KNN predicts salary category by looking at the most similar past cases and choosing the salary group that those neighbors belong to.

Why It's Useful

- Captures nonlinear patterns in salary data
- Compares each job record to its closest neighbors
- Provides contrast to linear models (Logistic Regression)
- Helps test whether local similarity predicts salary category

Key Takeaway

- KNN identifies some nonlinear salary patterns but is sensitive to scaling and noise, resulting in moderate performance compared to ensemble models.

Why This Matters

- Shows salary patterns are **not purely linear**
- Confirms that **local role similarity** helps but isn't enough
- Sets the stage for more powerful nonlinear models (SVM, ensembles)

Metric	Score
Accuracy	64.71%
Precision	67.31%
Recall	71.33%
F1 Score	69.26%
ROC-AUC	68.37%

Support Vector Machines

What is it?

Support Vector Machines classify salary groups by finding the best boundary that separates higher-salary and lower-salary cases.

- **Linear SVM** uses a straight-line boundary.
- **RBF SVM** uses a curved boundary that adapts to complex patterns.
- **Polynomial SVM** uses a curved boundary shaped by polynomial interactions between features.

Why It's Useful

- SVMs are powerful when salary patterns aren't perfectly linear.
- **Linear SVM** works well when features separate cleanly.
- **RBF SVM** handles nonlinear relationships and subtle patterns.
- **Polynomial SVM** captures interactions between features (e.g., job title × experience) that influence salary together.

Key Takeaway

- Polynomial SVM adds flexibility by modeling **feature interactions**, making it more expressive than Linear SVM and sometimes more stable than RBF SVM when patterns follow structured curves.

Why This Matters

- Including Polynomial SVM shows how salary classification changes when the model considers **combined effects** of features — not just individual contributions. This helps reveal deeper salary patterns that simpler boundaries might miss.

Support Vector Machines Performance

Model	Accuracy	Precision	Recall	F1	AUC
Linear	63.39%	62.45%	86.07%	72.38%	68.70%
RBF	65.38%	67.92%	71.79%	69.80%	65.59%
Polynomial	65.36%	67.45%	73.16%	70.19%	66.31%

Ensemble Models

“High-performance models that combine many decision trees for stability and accuracy.”

Decision Tree

What is it?

A Decision Tree makes choices step-by-step, like a flowchart. It splits the data into groups based on factors such as job title and experience

Why It's Useful

- Provides a simple, interpretable nonlinear model
- Shows how features split the population into salary groups
- Serves as the foundation for all ensemble methods
- Helps visualize hierarchical salary decision patterns
- Sets the stage for Random Forest, Gradient Boosting, and Extra Trees

Key Takeaway

- Decision Trees capture nonlinear salary relationships but tend to overfit, making them less stable than ensemble models

Why This Matters

- Shows ensemble averaging boosts reliability
- Demonstrates strong nonlinear modeling capability
- Provides a benchmark for more advanced ensembles

Metric	Score
Accuracy	65.26%
Precision	67.74%
Recall	71.94%
F1 Score	69.78%
ROC-AUC	68.95%

Random Forest

What is it?

Random Forest uses many decision trees and combines their answers. Think of it like getting multiple opinions instead of relying on just one.

Why It's Useful

- More stable and reliable than a single tree
- Handles messy or complex data well, which is true in this case
- Provides stable feature importance rankings
- Strong baseline ensemble model for tabular data

Key Takeaway

- Random Forest improves stability and accuracy over single trees, offering strong performance across all metrics.

Why This Matters

- Shows ensemble averaging boosts reliability
- Demonstrates strong nonlinear modeling capability
- Provides a benchmark for more advanced ensembles

Metric	Score
Accuracy	65.25%
Precision	67.62%
Recall	72.25%
F1 Score	69.86%
ROC-AUC	69.06%

Gradient Boosting

What is it?

Gradient Boosting builds trees one at a time, each one fixing the mistakes of the previous one. It's like learning from errors step-by-step.

Why It's Useful

- Sequentially reduces bias by correcting previous errors
- Excellent for capturing subtle patterns in tabular data
- Often very accurate on structured business data
- Provides a strong comparison point against Random Forest and Extra Trees

Key Takeaway

- Gradient Boosting uncovers complex salary interactions but may require tuning to reach peak performance.

Why This Matters

- Shows the value of sequential learning
- Highlights deeper nonlinear structure in salary data
- Sets up the comparison with Extra Trees

Metric	Score
Accuracy	65.24%
Precision	67.71%
Recall	71.94%
F1 Score	69.76%
ROC-AUC	69.00%

AdaBoost

What is it?

AdaBoost pays extra attention to the records that are hardest to classify. It tries to “boost” accuracy by focusing on the tricky cases.

Why It's Useful

- Focuses on misclassified samples
- Useful for datasets with overlapping salary classes
- Adds diversity to your ensemble comparison
- Helps test whether focusing on difficult examples improves results

Key Takeaway

- AdaBoost improves performance on difficult cases but can be sensitive to noise, making it less stable than Random Forest and Extra Trees.

Why This Matters

- Shows how reweighting difficult cases affects salary prediction
- Demonstrates strengths and limitations of boosting
- Provides contrast to Gradient Boosting and Extra Trees

Metric	Score
Accuracy	65.51%
Precision	67.69%
Recall	72.94%
F1 Score	70.22%
ROC-AUC	69.29%

Extra Trees

What is it?

Extra Trees builds many highly random trees and averages their results. This randomness actually makes the model more stable and accurate.

Why It's Useful

- Uses randomized splits for faster, more robust learning
- Often outperforms Random Forest and is one of the best models for business data
- Works extremely well with SHAP for explaining results
- Ideal for datasets with many categorical variables
- Handles lots of categories and one-hot encoded features well

Key Takeaway

- Extra Trees delivered the strongest overall performance, offering the best balance of accuracy, stability, and interpretability

Why This Matters

- Confirms ensemble methods are best for salary classification
- Provides the most reliable model for downstream SHAP analysis
- Forms the foundation for your final recommendation

Metric	Score
Accuracy	64.89%
Precision	67.40%
Recall	71.67%
F1 Score	69.47%
ROC-AUC	68.38%

Which factors matter most in predicting salary level?

- The model’s decisions are driven mostly by job role, experience, and company characteristics — the same factors compensation teams use in real-world salary decisions.

Rank	Model Variable	What does this mean?
1	Experience Level	Experience level is the strongest driver of salary — senior and executive roles push pay upward the most.
2	Job Title Group	Job type heavily influences compensation — analytics, ML/AI, and data science roles command higher salaries.
3	Company Size	Larger companies tend to offer higher salary bands; small companies show lower base pay.
4	Employment Type	Full-time roles pay more; part-time and contract roles reduce predicted salary.
5	Employee Remote Work	Remote vs. hybrid vs. on-site work affects salary bands depending on company policy and location flexibility.
6	Work year	More recent work years reflect updated market salary trends — newer years push salary predictions upward.
7	Employee Residence	Where the employee lives affects salary — U.S. residence increases predicted pay; other countries lower it.

How each feature pushes salary predictions up or down.

- Insights:
 - SHAP tells us how much each feature influences the model's decision. Think of it as showing which factors push a salary prediction higher or lower.
 - It's the closest thing to "opening the model's brain" and seeing how it thinks.
- Key Takeaway:
 - The model's decisions make intuitive sense: job role, experience, company size, and remote status are the strongest drivers of salary predictions — exactly what compensation teams expect.

Rank	Variable	Interpretation
1	Experience Level	Strongest predictor of salary. Senior and Executive categories dominant.
2	Job Title Group	ML/AI Engineering, Other Tech, Data Science, and Analytics roles drive salary upward.
3	Remote Work	Remote vs. On-Site meaningfully shifts salary predictions.
4	Work Year	Newer work years reflect updated market salary trends.
5	Employment Type	Full-time roles predict higher salary; part-time lowers it.
6	Company Size	Medium and small companies show lower SHAP impact compared to experience and job type.
7	Employee Residence	Location has the smallest SHAP impact in your reduced model.

Choosing the “Best” Model & Final Thoughts Summary

Selecting the best model

Criteria



- **How We Selected the Best Model**
 - We compared the final classification machine learning models side-by-side using the same five measures. The goal was to find the model that is:
 - Most accurate overall
 - Most consistent across metrics
 - Best at identifying high-salary cases
 - Most reliable for real-world use

Classification Model Comparison

Model Type	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	AUC (%)
Linear & Probabilistic Models	Logistic Regression	65.27	67.39	73.01	70.09	69.08
	Gaussian NB	56.16	56.01	99.45	71.66	65.67
	Bernoulli NB	61.84	65.46	66.75	66.10	66.29
Distance & Margin-Based Models	Linear SVM	63.39	62.45	86.07	72.38	68.70
	RBF SVM	65.38	67.92	71.79	69.80	65.59
	Polynomial SVM	65.36	67.45	73.16	70.19	66.31
	KNN	64.71	67.31	71.33	69.26	68.37
Ensemble Tree-Based Models	Decision Tree	65.26	67.74	71.94	69.78	68.95
	Random Forest	65.25	67.62	72.25	69.86	69.06
	Extra Trees	65.24	67.71	71.94	69.76	69.00
	Gradient Boosting (BEST)	65.51	67.69	72.94	70.22	69.29
	AdaBoost	64.89	67.40	71.67	69.47	68.38

Best Model Selection

Best Performing Model:

- Gradient Boosting

Why Gradient Boosting Stands Out:

- It had the highest overall accuracy
- It produced the strongest balanced score (F1)
- It captured high-salary cases than other models
- It remained stable and reliable across all measures
- It works extremely well with SHAP explanations, making results easy to interpret

Gradient Boosting is the most effective model for predicting whether a salary is above or below the median using a simplified feature set.

Model Limitations



Cannot explain *why* salaries differ beyond the data provided

It only uses the features in the dataset. Anything not included — negotiation, bonuses, internal policies — is not captured.



Cannot guarantee perfect accuracy

Even the best model makes mistakes. Salary data is complex, and some cases will always be hard to classify.



Cannot account for future market changes

Shifts in hiring trends, remote work policies, or economic conditions may change salary patterns over time.



If the dataset reflects existing market biases, the model may unintentionally learn and reproduce them.



Key Takeaway: The model is useful for understanding salary patterns and supporting decisions, but it is a decision-support tool, not a decision-making tool. Compensation teams still need to interpret results in context.

Next Steps

Expand the Dataset With More Demographics & Company Details

- Collect richer information such as job-level attributes, education, industry, benefits, and additional compensation factors. More detailed data → stronger, more accurate models.

Build a Data Pipeline That Automatically Cleans and Prepares New Data

- Create a workflow that: 1. gathers new salary records, 2. cleans and standardizes them, 3. updates encoded features, 4. prepares the dataset for modeling, and 5. This ensures the model always works with fresh, high-quality data.

Automate Model Training When New Rows Arrive

- Set up the pipeline so the model **re-trains itself** whenever new data is added. This keeps predictions aligned with current market trends.

Use the Provided Models as the Starting Point

- Begin with the models tested in this project. Start with **Gradient Boosting**, since it performed best in the reduced set.

Select the Best Model Using Clear Criteria

- **F1 Score** (balanced performance)
- **ROC-AUC** (ability to separate salary groups)

Key Takeaway

- The next step is building a full, automated pipeline that continuously collects data, cleans it, retrain models, and selects the best performer using F1 and ROC-AUC

Final Summary



The reduced feature set is strong and reliable

Even with fewer inputs, the models performed consistently and captured meaningful salary patterns.



Gradient Boosting is the best-performing model

It delivered the strongest balance of **F1 Score** and **ROC-AUC**, the two criteria used for model selection.



Salary predictions align with real-world compensation logic

Job role, experience level, company size, and remote status were the most influential factors — exactly what compensation teams expect.



SHAP results make the model transparent and explainable

We can clearly see how each feature pushes predictions up or down, supporting trust and interpretability.



A future pipeline can automate everything

Collect richer data, clean it automatically, retrain models as new rows arrive, and select the best model using F1 + ROC-AUC.



Key Takeaway: This project demonstrates a stable reduced model, clear salary insights, and a path toward a fully automated, continuously improving salary prediction pipeline.